ModelIt!

Teacher's Guide

The Crime Scene in Every Rock

Reading Earth's History One Layer at a Time

Quick Reference

Grade Level	Grade 7
NGSS Standard	MS-ESS1-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-ESS1-4

Construct a scientific explanation based on evidence from rock strata for how the geologic time scale is used to organize Earth's 4.6-billion-year-old history.

Three-Dimensional Learning

Science Practice	**Constructing Explanations** — Students construct scientific explanations using evidence from rock strata to explain how geologists organize Earth's history.
Disciplinary Core Idea	**ESS1.C The History of Planet Earth** — Rock strata and the fossil record can be used as evidence to organize the relative occurrence of major historical events in Earth's history.
Crosscutting Concept	**Scale, Proportion, and Quantity** — Students work with vast timescales (millions to billions of years) and understand how thin layers of rock can represent enormous spans of geological time.

CAST Testing Alignment

MULTIPLE CHOICE:

A geologist finds a rock layer containing marine fossils between two layers containing land plant fossils. Which interpretation is best supported by this evidence?

CONSTRUCTED RESPONSE:

Examine a diagram of rock strata with different fossils in each layer. Using the principle of superposition and your knowledge of the fossil record, construct an explanation of the geological history this sequence represents.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Earth is 4.6 billion years old, and its history is written in stone — literally. Rock strata (layers) form a record of environmental conditions, life forms, and geological events that have shaped our planet over deep time.

How Rock Layers Form

Sedimentary rock forms when particles (sand, mud, shells, organic material) settle and accumulate in layers. Over millions of years, pressure from overlying layers compresses and cements these sediments into solid rock. Each layer represents a specific time period and environment.

Principle of Superposition

In undisturbed rock sequences, the oldest layers are at the bottom and the youngest are at the top. This principle, established by Nicolas Steno in 1669, allows geologists to determine the RELATIVE age of rock layers and the fossils they contain without needing absolute dates.

Fossils as Evidence

Fossils are the preserved remains or traces of ancient organisms. They form when organisms are rapidly buried in sediment before decomposition. Index fossils — from organisms that lived for a short time but across a wide geographic area — are especially useful for dating and correlating rock layers across distant locations.

Gaps in the Record

Unconformities are gaps in the rock record where erosion has removed layers or deposition didn't occur. These represent missing time — potentially millions of years of history erased. Geologists identify three types: angular unconformities, disconformities, and nonconformities.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — **Locate the System**
Students identify sedimentation rate, erosion rate, rock layer thickness, and fossil preservation quality as components of the geological record system.

E — **Establish Relationships**
Students determine that sedimentation builds layers while erosion destroys them, and that rapid burial improves fossil preservation.

V — **Visualize & Model**
Students build a model showing how sedimentation and erosion interact to create (or destroy) the rock record.

E — **Evaluate Outcomes**
Students run rapid deposition, heavy erosion, and balanced scenarios to observe different geological outcomes.

R — **Revise & Extend**
Students add tectonic activity or sea level change to explore how geological forces alter the record after formation.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Every rock layer is a page in Earth's history book. Let's learn to read it."*

Watch For

Show photos of Grand Canyon layers. Each color = different time period.

Slide 2: Learning Objectives

3 min

Say: *"Today you're becoming geological detectives — reading clues written in stone."*

Watch For

Have students read objectives. Pre-teach 'stratigraphy' and 'superposition.'

Slide 3: Big Question

3 min

Say: *"If I hand you a rock with a fish fossil in it, what can you figure out?"*

Watch For

Pass around fossil samples. Students guess: marine, freshwater, or land?

Slide 4: LEVER Framework

2 min

Say: *"We'll model how the rock record is created — and sometimes destroyed."*

Watch For

Preview LEVER steps.

Slide 5: Activity 1: Components

8 min

Say: *"What builds layers? What destroys them? What determines fossil quality?"*

Watch For

Guide sorting. Discuss why sedimentation and erosion are external processes.

Slide 6: Activity 2: Connections

8 min

Say: *"Sedimentation builds the record. Erosion erases it. What wins?"*

Watch For

Help students see the competing processes.

Slide 7: Activity 3: Simulation

10 min

Say: *"What happens when a massive flood buries everything vs. when erosion dominates?"*

Watch For

Students predict first. Compare scenarios.

Slide 8: Discoveries

5 min

Say: *"Why is the fossil record incomplete? What creates gaps in Earth's story?"*

Watch For

Discuss unconformities. Show real examples of missing time.

Slide 9: STEM Challenge

12 min

Say: *"Build Earth's history in a jar — each layer tells a different story."*

Watch For

Groups build layered core samples with different materials and 'fossils.'

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Sedimentation Rate, Erosion Rate
- **INTERNAL:** Rock Layer Thickness, Fossil Preservation Quality

Relationships

- **Sedimentation Rate to Rock Layer Thickness:** POSITIVE (+) — Faster sedimentation means more material is deposited, creating thicker rock layers over time.
- **Erosion Rate to Rock Layer Thickness:** NEGATIVE (-) — Higher erosion rates wear away existing layers, reducing thickness and creating gaps (unconformities) in the rock record.
- **Rock Layer Thickness to Fossil Preservation Quality:** POSITIVE (+) — Thicker, more rapidly deposited layers bury organisms quickly, protecting them from decomposition and creating better-preserved fossils.

Key Discoveries

- Rock layers are like pages in a book — each one records conditions at the time it formed
- Rapid burial preserves fossils best because organisms are protected from decomposition and scavengers
- Erosion creates gaps in the rock record — millions of years of history can be missing from a single location
- The principle of superposition allows us to determine which layers (and fossils) are older or younger

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Rocks don't change"	Rocks are constantly being created, destroyed, and recycled through the rock cycle. Sedimentary rocks can be metamorphosed by heat and pressure, melted into magma, or eroded into sediment that forms new rock. Earth's surface is dynamic.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Fossils are just pretty rocks"	Fossils are preserved evidence of ancient life — they contain critical information about what organisms existed, what the environment was like, and how life has changed over time. Each fossil is a data point in the story of evolution.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The Earth has always looked like it does now"	Earth's surface has changed dramatically over 4.6 billion years. Continents have moved, oceans have appeared and disappeared, mountains have risen and eroded away. The landscape you see today is just a snapshot of an ever-changing planet.